



SmartFitao AI

Tailor Crafted · Virtually Try-On

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Abstract

No international platform supports tailor customization, LLC registration cost, or brand connections. Users lack order tracking security and cannot try cloths at home. SmartFitao AI solves this with a 3D Marketplace featuring Order Tracking, 3D Studio for tailors, Kameez Size Prediction (age/height/weight), 33-landmark A-Pose detection, 2D Virtual Try-On, and AI Chatbot for 3D product recommendation.

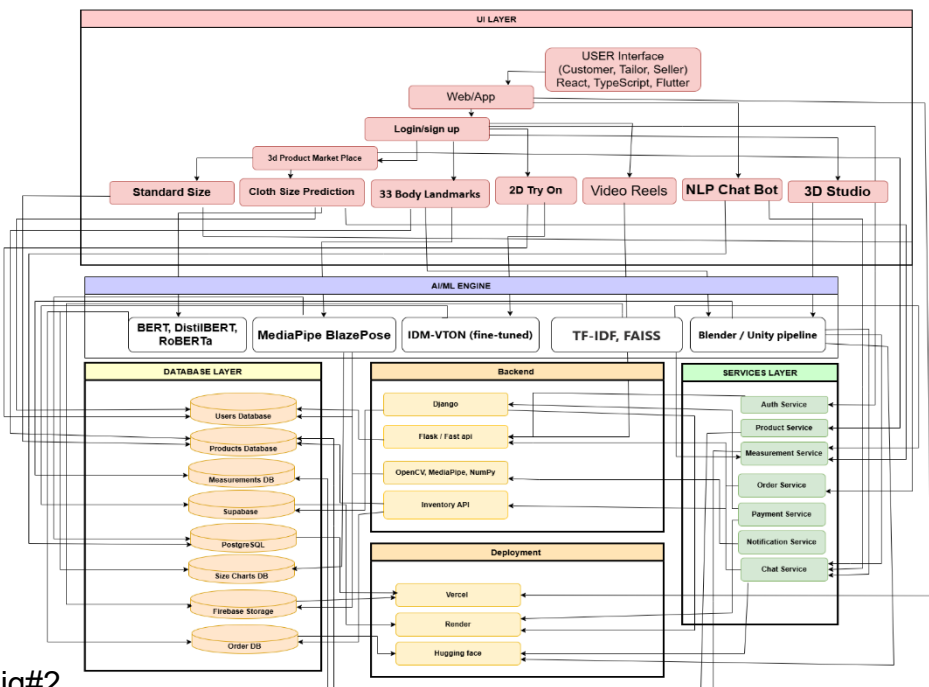


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Fig#1: Logo

Technical Architecture



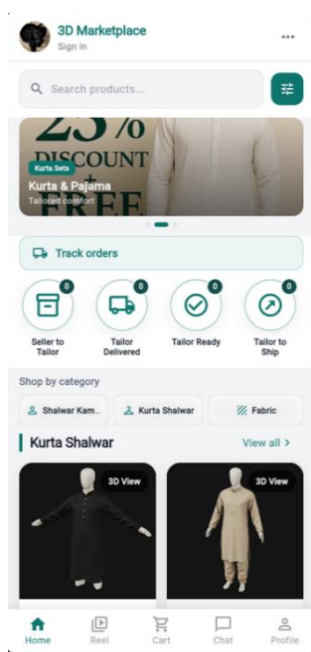
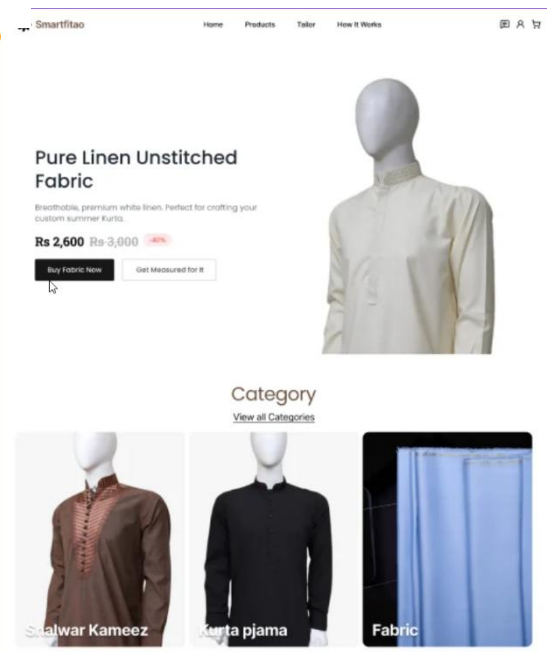
Fig#2

Introduction

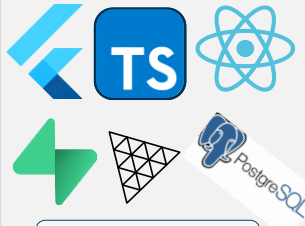
In SmartFitao AI, we surveyed **14 tailor** shops to identify real problems: lack of connection with **international clients**, no platform supports tailor customization and brand connections. It provides users with order tracking security and the ability to try cloths at home. The platform features a 3D Marketplace with Order Tracking, 3D Studio for tailors, Cloth Size Prediction (age/height/weight), 33-landmark A-Pose detection, 2D Kurta Virtual Try-On, and **3D Product Recommendation AI Chatbot**. Built with Django, React, Flutter.

Methods and Materials

- Cloth Size Prediction Model:** Train a model that predicts kameez size using age, height, weight and custom questions.
- Make Kurta Dataset:** With BERT & Random Forest on 50,000 body rows from 14 tailor surveys + 1,200 cloth sizes
- 33 Body Landmarks:** Computer Vision using MediaPipe, BlazePose for capture
- 2D Virtual Try-On:** Fine-tuned IDM-VTON to work for Kurta and Kameez try-on
- 3D Product Recommendation Chatbot:** TF-IDF, FAISS & Groq Llama for product recommendations in 3D visualization way



3D Market Place (Web, App)



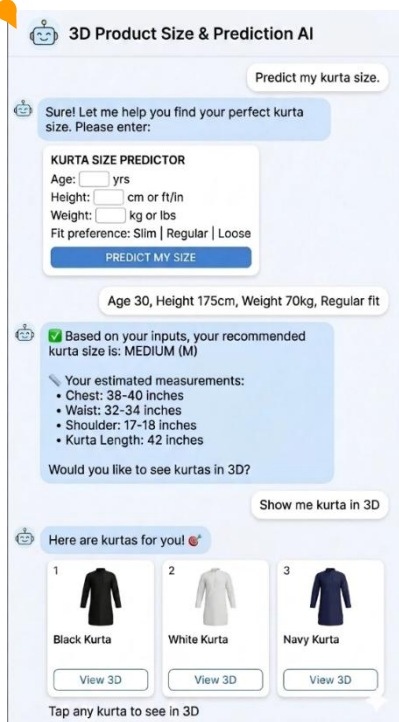
Fig#3

Discussion

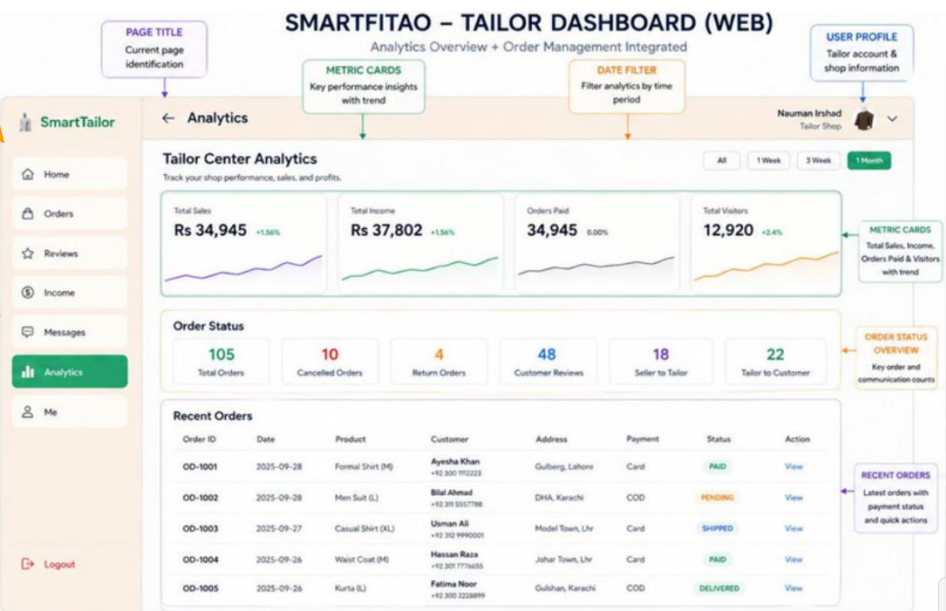
- Continuing work on PIFuHD metaverse model for 3D body try-on creation from single picture, fine-tuning to capture both back and front sides with JSON files to create 3D base body, 3D body try-on remains in progress..



3D Tailor Studio Fig#5



AI Chatbot Fig#4



Fig#6

Results

Model / Feature	Accuracy
Cloth Size Prediction (BERT)	87.6%
NLP 3D Chatbot Answer Match	87.84%
33 Landmark Detection	94%
2D Virtual Try-On	91%

Table#1

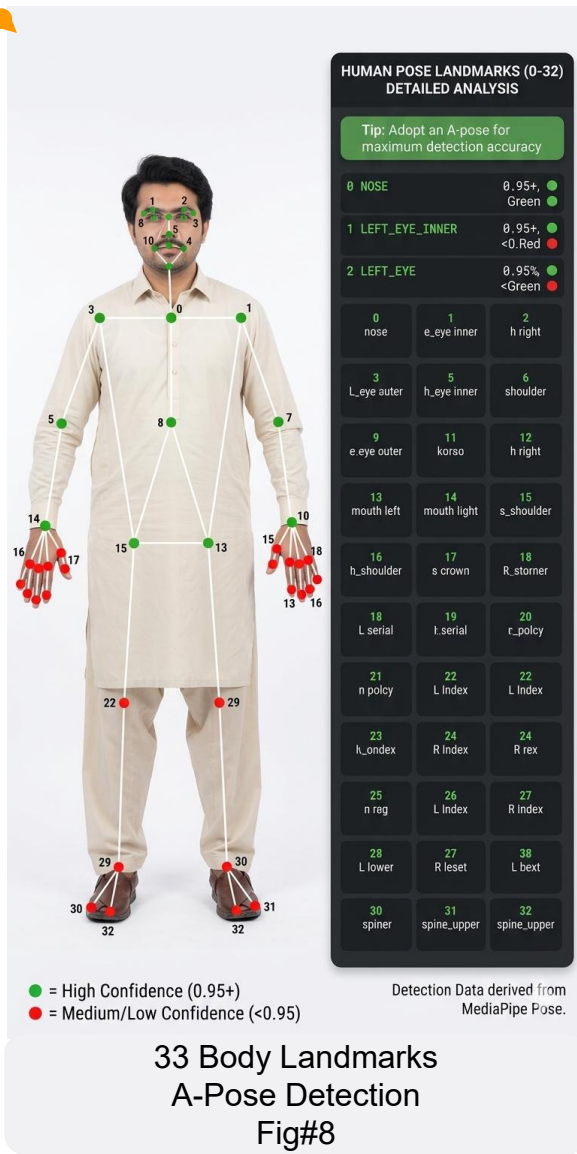
Feature	Test Case	Defects	Correct
Web/App Login Auth	TC-1	1	1
AI Cloth Prediction	TC-2	1	1
33 Landmark Calculation Accuracy	TC-3	1	1
2D Virtual Try-On	TC-4	2	2
3D Chatbot Response	TC-5	2	2
3D Product Rander	TC-6	6	6

References

- Choi et al. (2024).** IDM-VTON. *arXiv:2403.05139*
- Nashita et al. (2024).** Cloth Size Prediction using Random Forest. *IEEE (10.1109/ICSSECC12345)*
- Gondal et al. (2025).** RAG + FAISS for Fashion Chatbot. *arXiv:2511.19149*

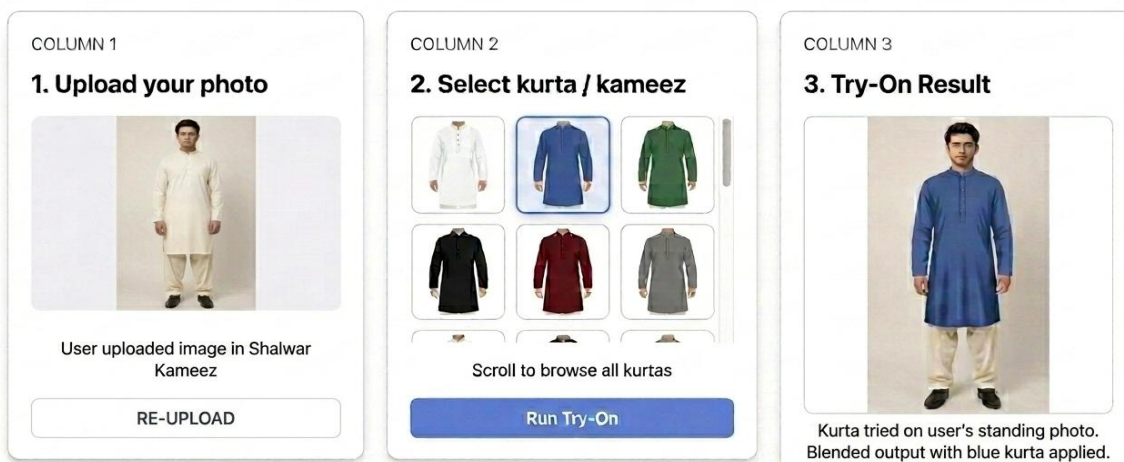


Size Prediction Model Kurta Dataset Fig#7



33 Body Landmarks A-Pose Detection Fig#8

SmartFitao - 2D SHALWAR KAMEEZ TRY ON



IDM-VTON Fine-Tuned Fig#9

Conclusion

SmartFitao AI **successfully** bridged the gap between international customers and Pakistani local tailors.

- 2D Virtual Try-On** for Kurta and Kameez at home
- Kameez Size Prediction** from user age, height, and
- 3D Product Recommendation Chatbot**